SAT-EURS Track 2: Embedded Edge Champion Architecture

track2-apex-orin-v3 (Official Submission #19447 | Leaderboard Composite: 0.407)

Track: Track 2 (Jetson / Edge) | **Target:** NVIDIA Jetson AGX Orin | **Date:** October 28, 2026 | **Team:** SAT-EURS (MIT-BLR)

1. Executive Summary & Jetson AGX Orin Compliance

track2-apex-orin-v3 is the official champion submission for EURS Track 2, scoring **0.407 Composite** on the competition server. Designed for real-time edge execution on the **NVIDIA Jetson AGX Orin**, it satisfies all strict hardware limits with significant safety margins: **Params ≤ 1.0 M**, **Compute ≤ 12.0 GFLOPs**, and **100% ONNX Whitelist Operator Compliance**.

Metric / Parameter	Official Cap	EURS-Net V5 Baseline (#18699)	track2-apex-orin-v3 (#19447)	Status & Delta
Composite Score	Highest Wins	0.338	**0.407**	**+20.4% Official Server Surge**
Total Parameter Count	≤ 1,000,000 (1.0 M)	768,300 (0.768 M)	**884,120 (0.884 M)**	PASSED (11.6% Headroom)
Computational Complexity	≤ 12.00 GFLOPs	9.050 GFLOPs	**9.820 GFLOPs**	PASSED (18.2% Headroom)
Full-Ref PSNR (dB)	Normalized 15-35 dB	17.10 dB (blurry)	**23.25 dB**	**+6.15 dB Restoration Leap**
Full-Ref SSIM	Higher is better	0.821	**0.847**	+0.026 Structural Gain
Segmentation Q_{seg}	mIoU × F1 × Cov	0.245	**0.227** (mIoU: 0.406 / F1: 0.560)	Zero false-positive elasmobranch collapse
ONNX Whitelist Compliance	Zero Non-Whitelist	Compliant	**100% Certified Compliant**	Zero unapproved custom operators

2. 18-D Radiometric Condition Router & Physical Basis Contraction

Standard edge baselines spend 5.8 GFLOPs on 2D convolutional restoration branches that output blurry images (17.1 dB PSNR). `track2-apex-orin-v3` replaces this with a mathematical breakthrough that costs **less than 0.30 GFLOPs** while boosting PSNR to **23.25 dB**:

- **18-D Geometry-Invariant Radiometric Vector $s(x)$:** Extracted on a 256×256 pooled representation: (1) 3 channel means (μ_R, μ_G, μ_B), (2) 3 channel standard deviations $(\sigma_R, \sigma_G, \sigma_B)$, (3) 3 soft-mins and 3 soft-maxs, (4) 3 differential attenuation differences $(\mu_B - \mu_R, \mu_G - \mu_R, \mu_B - \mu_G)$, and (5) horizontal and vertical spatial gradient magnitudes.
- **Direct Condition Router Trunk:** A lightweight 2-layer LayerNorm MLP with OnnxSafeHardswish activations maps $s(x) \in \mathbb{R}^{18}$ to a 32-D condition embedding e_{cond} and 200 condition logits.
- **Scale-Consistent Matrix Bank (`tbl19_20sc_scale_consistent.pt`):** Evaluates matrix contraction over a precomputed static buffer $[200, 57]$ (10 water types $\times$ 20 depth levels from 5m to 100m) to synthesize $W_{\text{eff}} \in \mathbb{R}^{19 \times 3}$, multiplying the 19-term physical basis at full 1024×1024 resolution in real time.

3. FiLM Spatial Refiner & In-Graph Gaussian Unsharp Filter

To break the linear polynomial ceiling without consuming heavy FLOPs, a specialized spatial residual refiner operates at downsampled 256×256 resolution:

- **FiLM-Modulated Inverted Residual Blocks:** 3 depthwise-separable residual blocks (32 channels) modulated by the condition embedding: $\text{feat} = \text{feat} \times (1.0 + 0.5 \tanh(\gamma)) + 0.5 \tanh(\beta)$. Zero-initialized weights guarantee that step 0 matches the exact physical RSUIGM output.
- **Bounded Residual:** Predicts $\Delta I_{256} = 0.25 \cdot \tanh(\text{Conv}(\text{feat}))$, bilinearly upsampled to 1024×1024 and added to the physical reconstruction.
- **In-Graph Depthwise Gaussian Unsharp Masking:** Convolves restored image with a normalized 3×3 Gaussian blur kernel $K = \frac{1}{16}[1, 2, 1; 2, 4, 2; 1, 2, 1]$ and blends high-frequency gradients: $I_{\text{sharp}} = \text{clip}(I + 0.16 \cdot (I - I * K), 0, 1)$, pushing official SSIM to 0.847.

4. Truncated MobileNetV3-Large & Apex Pyramid Decoder

The segmentation subnet captures multi-scale semantic context while respecting the 1.0M parameter ceiling:

Subnet Module	Architectural Design	Parameters / FLOPs
Truncated MobileNetV3 Encoder	Truncated strictly at **Block 11 (`features[:12]`)**. Captures Stride 4 (24ch), Stride 8 (40ch), and Stride 16 (112ch) features. Cutting Block 12 saves 386,880 parameters.	**406,368 params** Pretrained ImageNet
Dual-Input RGB Stem	Auxiliary stem ingests raw degraded RGB (3×3 Stride-2 conv, 16ch), fused into restored RGB stem with zero-initialized BatchNorm γ to preserve pretrained filters.	**~1,200 params** Zero initial drift

Stride-32 Dilated ASPP Stage	Custom context downsampler (112 $\to$ 128ch) with parallel dilated branches ($d=1, 2, 4$ and global pooling) modulated by FiLM condition gate $c_{\text{mod}} = 1.0 + 0.3\tanh(W e_{\text{cond}})$.	**~115,000 params** Wide spatial field
Progressive 4-Scale FPN	72-channel lateral skip projections with progressive top-down depthwise-separable fusion blocks ($S32 \to S16 \to S8 \to S4$). Final 1×1 conv predicts 5 class logits.	**~220,000 params** Continuous borders
Bounded ScenePriorGate	Global scene descriptor g_{32} and condition vector predict bounded logit bias $\Delta z = 0.35 \cdot \tanh(\text{MLP}([g_{32}, e_{\text{cond}}]))$, dynamically preventing elasmobranch hallucinations on reef scenes.	**~12,500 params** Hallucination gate
Spatial Speckle Suppressor	Embedded in-graph AvgPool2d(11, stride=1, padding=5) evaluates cluster support for rare classes 3 & 4. Applies penalty $\frac{1.2}{0.04}\text{ReLU}(0.04 - \text{support})$ to eliminate isolated single-pixel false alarms.	**0 params** 0.01 GFLOPs

5. ONNX Whitelist Operator Replacement & Export Audit

To guarantee native compilation on NVIDIA Jetson TensorRT and ONNX Runtime with zero custom operator dependencies, all unsupported activations were replaced with mathematical equivalents constructed from basic ops:

• nn.Hardswish(x) → x * F.relu6(x + 3.0) / 6.0 (strictly elementary Mul, Add, ReLU6, Div)

• nn.Hardsigmoid(x) → F.relu6(x + 3.0) / 6.0

• **Compiled Graph:** Single standalone model.onnx (3.18 MB) deployed inside deploy/track2/submission_track2_apex_orin_v3.zip. Static input shape: $[1, 3, 1024, 1024]$; Outputs: restored_image $[1, 3, 1024, 1024]$ and seg_logits $[1, 5, 1024, 1024]$.